Magnetic quotients and a restricted global semicircle law

WoW research programme

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Status and scope

This is a complete project derivation, not a claim of priority or an externally refereed theorem. It concerns the **global empirical spectral measure of a compressed observable**, not microscopic BGS statistics of a quantum propagator. The arithmetic restriction is strong: prime subgroup orders $K = o(\log p)$. Section 9 proves that this regime is incompatible with the orders of any one fixed hyperbolic integral cat map. The exact operator identities and computational representation do not require that restriction.

The argument uses an explicit finite Heisenberg-group product, not the historical unverified word-rewrite claims. Independent implementations check its finite algebra; they are not substitutes for the limiting proof.

1. The exact matrix being studied

Let p be an odd prime, let $H \le \mathbb{F}_p^*$ have odd order K, and let $q \in \mathbb{F}_p^*$. All exponents and fractions inside $e_p(t) = \exp(2\pi i t/p)$ are interpreted in $\mathbb{F}_p$. On $\mathbb{C}^{\mathbb{F}_p}$, with the counting-measure inner product, define

$$W(a,b)f(x) = e_p(a(x-b/2))f(x-b), \qquad W_q = W(q,1),$$

$$S_h f(x) = f(h^{-1}x), \quad Q = \frac{1}{K}\sum_{h\in H} S_h, \quad R = S_{-1}, \quad P_- = (I-R)/2.$$

The projections Q, P_- commute. Set

$$\mathcal{E} = \operatorname{ran}(QP_-), \quad n = \dim \mathcal{E} = \frac{p-1}{2K}, \qquad B_q = W_q + W_q^*,$$

$$A_q = (QB_qQ)|_{\mathcal{E}}, \qquad X_q = \sqrt{K/2}\, A_q, \qquad \mu_m(q) = \frac{1}{n}\operatorname{Tr}_{\mathcal{E}} X_q^m. \tag{1}$$

The dimension follows by using signed indicators of the nonzero orbits of $H \cup (-H)$; odd functions vanish at zero. The identity $RW_qR = W_q^*$ shows that B_q preserves parity. Thus A_q is Hermitian and $\|A_q\| \le 2$. The notation A_q here is an operator, not the Cayley set in other notes.

For each odd multiplicative character χ trivial on H, extend $f_\chi(x) = \chi(x)/\sqrt{p-1}$ by zero at $x = 0$. These form an orthonormal basis of $\mathcal{E}$. Direct substitution gives

$$M_{\chi,\psi} = \frac{e_p(-q/2)}{p-1}\sum_{x\neq 0,1} \overline{\chi(x)}\psi(x-1)e_p(qx), \qquad [A_q] = M + M^*. \tag{2}$$

This is precisely the centered full-complex matrix in the maintained `spectral_diagnostics.py`, not a surrogate ensemble. Compression of $RW_qR = W_q^*$ to one parity sector also gives $M = M^*$; it does not make M entrywise real.

2. An explicit antiunitary and an isospectral real gauge

Define

$$(F_q f)(x) = p^{-1/2} \sum_y e_p(qxy) f(y), \qquad (Cf)(x) = \overline{f(x)}, \qquad \Theta_q = F_q C.$$

Proposition 1. The map Θ_q is an antiunitary involution preserving $\mathcal{E}$ and commuting with A_q. For a nontrivial character,

$$\Theta_q f_\chi = a_\chi(q) f_\chi, \qquad a_\chi(q) = \frac{\chi(q)\tau(\overline{\chi})}{\sqrt{p}}, \qquad \tau(\overline{\chi}) = \sum_{y \neq 0} \overline{\chi(y)} e_p(y), \quad |a_\chi(q)| = 1. \tag{3}$$

Choose any square roots $d_\chi^2 = a_\chi(q)$ and put $D = \mathrm{diag}(d_\chi)$. Then $D^*[A_q]D$ is real symmetric and has exactly the spectrum of A_q.

Proof. Fourier orthogonality gives $F_q\overline{F_q} = I$. Substitution $y = z + 1$ in $\Theta_q W_q$ gives

$$e_p(qxy - q(y - 1/2)) = e_p(q(x - 1/2)) e_p(q(x - 1)z),$$

which proves $\Theta_q W_q = W_q \Theta_q$. Changing variables in the Fourier sum yields $\Theta_q S_h = S_{h^{-1}} \Theta_q$, hence invariance of Q, R. For $x \neq 0$, substitution $u = qxy$ proves (3); at zero the character sum vanishes. Expanding the square of the Gauss sum and using orthogonality gives $|\tau(\chi)|^2 = p$. Finally $\Theta_q(d_\chi f_\chi) = d_\chi f_\chi$. A commuting operator has real entries in this fixed basis, and Hermiticity makes them symmetric. □

This is a geometric conjugation, not conjugation chosen after diagonalizing an arbitrary Hermitian matrix. Nonetheless an antiunitary symmetry alone is not a GOE universality theorem. In particular, the old operation of taking $[A_q]$.real is not reinstated: it is generally not isospectral.

3. Orbit assembly with only p minus two contributions

Choose representatives r_i of $\mathbb{F}_p^*/(H \cup -H)$. Let $i(x)$ be the orbit index, and let $s(x)$ be $+1$ on Hr_i and -1 on $-Hr_i$. The orbit basis is $v_i(x) = s(x)/\sqrt{2K}$ on its orbit. Consequently

$$[M]_{ij}^{\mathrm{orb}} = \frac{1}{2K} \sum_{\substack{x \neq 0,1 \\ i(x)=i,\ i(x-1)=j}} s(x)s(x-1)e_p(q(x-1/2)), \qquad [A_q]^{\mathrm{orb}} = M^{\mathrm{orb}} + (M^{\mathrm{orb}})^*. \tag{4}$$

There are $p - 2$ contributions, so sparse arithmetic assembly costs $O(p)$ and there are at most $O(p)$ nonzero locations before coalescing. This avoids a naive $O(n^2 p)$ character-sum construction. The reference implementation returns a **dense** $n \times n$ matrix, so its storage is $O(n^2)$; sparse assembly does not remove eigensolution costs. The overlap of the two orthonormal bases proves exact unitary equivalence.

4. All-order trace identity with every word case included

The Weyl and dilation rules are

$$W(a,b)W(c,d) = e_p((ad-bc)/2)W(a+c,b+d), \qquad S_g W(a,b) S_g^{-1} = W(a/g, gb). \tag{5}$$

Expand $(B_q Q)^m$ into words with $\epsilon_j \in \{-1,1\}$ and $h_j \in H$. Starting with $(g,a,b,c) = (1,0,0,0)$, update at each letter by

$$a' = a + \epsilon/g, \quad b' = b + \epsilon g, \quad c' = c + \epsilon(ag - b/g)/2, \quad g' = gh. \tag{6}$$

The product is $e_p(qc)W(qa,b)S_g$. Define

$$T_g(a,b) = \begin{cases} p\,\mathbf{1}_{a=b=0}, & g = 1, \\ e_p\Big(\dfrac{qab(1+g)}{2(1-g)}\Big), & g \neq 1. \end{cases} \tag{7}$$

Proposition 2. For every integer $m \geq 1$,

$$\operatorname{Tr}_{\mathcal{E}} A_q^m = \frac{1}{2K^m} \sum_{\epsilon_1,\dots,\epsilon_m;\ h_1,\dots,h_m} e_p(qc)\big(T_g(a,b) - T_{-g}(a,b)\big). \tag{8}$$

Proof. A diagonal entry of $W(qa,b)S_g$ requires $x = g^{-1}(x-b)$. For $g \neq 1$, its unique solution is $x = b/(1-g)$, giving (7). For $g = 1$, one needs $b = 0$; orthogonality of the remaining additive character gives the first line, since $q \neq 0$. Inserting R changes g to $-g$ and a, b to their negatives, preserving ab. The trace of the compressed power equals $\operatorname{Tr}(P_-(B_qQ)^m)$ by cyclicity and idempotence of Q. Expansion proves (8). Since H has odd order, $-g$ is never 1. No other cases are omitted. □

In particular there is an integer, possibly signed, histogram $C_m(t)$ such that

$$\operatorname{Tr} A_q^m = \frac{1}{2K^m} \sum_t C_m(t) e_p(qt), \qquad \mathbb{E}_{q\neq 0} \operatorname{Tr} A_q^m = \frac{pC_m(0) - \sum_t C_m(t)}{2K^m(p-1)}. \tag{9}$$

Histograms differing by a constant vector are equal at every nontrivial pth root. Negative coefficients cannot be discarded.

5. Reciprocal short-relation rigidity

Lemma 3. Suppose K is prime, $1 \leq m < K$, and

$$p^2 > m^{K-1}. \tag{10}$$

If $t_j \in H$ and $\epsilon_j \in \{-1,1\}$ satisfy

$$\sum_j \epsilon_j t_j = \sum_j \epsilon_j / t_j = 0 \quad \text{in } \mathbb{F}_p, \tag{11}$$

each label t occurs equally often with positive and negative sign. In particular m is even.

Proof. Write $t_j = h^{e_j}$, $0 \leq e_j < K$, and collect coefficients in $P(Z) = \sum_j \epsilon_j Z^{e_j}$. Its coefficient ℓ^1 norm is at most m. If $P(\zeta_K) \neq 0$, its norm N is a nonzero integer with

$$|N| = \prod_{a=1}^{K-1} |P(\zeta_K^a)| \leq m^{K-1}. \tag{12}$$

Multiplication by P in $\mathbb{Z}[Z]/\Phi_K$ has determinant N. Modulo p, Φ_K splits into distinct roots h^a, and (11) makes two distinct eigenvalues zero, at h, h^{-1}. Its nullity is at least two; Smith normal form then gives $p^2 \mid N$, contradicting (10)–(12). Thus $P(\zeta_K) = 0$. Irreducibility of Φ_K follows from Eisenstein applied after $Z \mapsto Z + 1$. Since $\deg P \le K - 1$, we have $P = c\Phi_K$ for an integer c. A nonzero c would give coefficient ℓ^1 norm at least $K > m$. Hence $P = 0$ coefficientwise, proving balance. $\square$

Both reciprocal equations and the strict cutoff are used. This sufficient cutoff is not an exact characterization of exceptional primes.

6. Main term and an explicit remainder

For words ending with $g = 1$, put $t_1 = 1$ and $t_j = h_1 \cdots h_{j-1}$ for $j \ge 2$. The labels $t_2, \dots, t_m$ are arbitrary in H, and the final h_m is determined. The state becomes

$$a = \sum_j \epsilon_j / t_j, \quad b = \sum_j \epsilon_j t_j, \quad c = \frac{1}{2} \sum_{i<j} \epsilon_i \epsilon_j (t_j/t_i - t_i/t_j).$$

Equations (1) and (8) give

$$\mu_m(q) = \frac{p}{p-1} \frac{1}{2^{m/2} K^{m/2-1}} \sum_{\substack{t_1=1,\ t_2,\dots,t_m \in H;\ \epsilon \\ a=b=0}} e_p(qc) + \mathcal{R}_m(q), \tag{13}$$

$$|\mathcal{R}_m(q)| \le \frac{2^{m/2+1} K^{m/2+1}}{p-1}. \tag{14}$$

Indeed every trace outside the $g = 1, a = b = 0$ contribution has modulus at most one; bounding the two traces of each of $(2K)^m$ words by two gives (14). Under Lemma 3 odd moments have no main term.

7. Pairings, overcounting, and crossing cancellation

Let $m = 2r$. Every balanced tuple admits a matching of the $2r$ positions, with equal labels and opposite signs on matched positions. Write

$$P_r = (2r-1)!!, \qquad C_r = \frac{1}{r+1} \binom{2r}{r}.$$

These are the numbers of all and of noncrossing matchings. Order pairs by their first endpoints, so pair 1 contains position 1. Assign labels $x_1 = 1, x_2, \dots, x_r \in H$ and orientations $\sigma_i \in \{-1, 1\}$. With distinct labels the matching is unique. Overcounting is possible only when two labels coincide. A union bound bounds the number of bad matching assignments by

$$P_r 2^r \binom{r}{2} K^{r-2}, \tag{15}$$

interpreted as zero for $r = 1$. Each balanced tuple has at least one matching; therefore (15) also bounds the excess multiplicity of the replacement sum.

Summing the four interactions between two matched pairs gives the exact Laurent polynomial

$$F_{\pi,\sigma}(x) = \sum_{\substack{i<j \\ \text{pairs } i,j \text{ cross}}} \sigma_i\sigma_j(x_j/x_i - x_i/x_j). \tag{16}$$

Disjoint or nested pairs give zero; interlaced pairs give the indicated term. Noncrossing matchings have $F = 0$. Any crossing matching has nonconstant F: each crossing contributes two unique directed-ratio monomials with opposite unit coefficients. After $x_1 = 1$, the monomials involving pair 1 become x_j, x_j^{-1} and remain distinct from every other term. Thus they cannot cancel in odd characteristic.

For a crossing let $d = r - 1$ and

$$S_F(q) = K^{-d} \sum_{x \in H^d} e_p(qF(x)).$$

Multiplication of $F(x) - F(y)$ by all $2d$ variables clears denominators and gives a nonzero polynomial of total degree at most $2d + 1 = 2r - 1$. The grid zero bound therefore gives at most $(2r-1)K^{2d-1}$ equal-value pairs. For completeness, that bound follows by induction on the number of variables: separate the leading coefficient in the last variable, use the univariate root bound off its zero set, and induct on that coefficient on the remaining fibres. Additive orthogonality now yields

$$\mathbb{E}_{q\neq 0}|S_F(q)|^2 = \frac{pK^{-2d}\#\{(x,y) : F(x) = F(y)\} - 1}{p-1} \le \frac{p(2r-1)}{(p-1)K}. \tag{17}$$

This uses no independence assumption for matrix entries, words, or crossings.

8. Quantitative theorem and limiting law

Theorem 4. Let K be an odd prime and let p be prime with $K \mid p-1$. For $2r < K$ and $p^2 > (2r)^{K-1}$, uniform nonzero q satisfies

$$\begin{aligned} \|\mu_{2r} - C_r\|_{L^2(q\neq 0)} \le{}& \frac{C_r}{p-1} \\ &+ \frac{p}{p-1}(P_r - C_r)\sqrt{\frac{p(2r-1)}{(p-1)K}} \\ &+ \frac{p}{p-1}\frac{P_r\binom{r}{2}}{K} + \frac{2^{r+1}K^{r+1}}{p-1}. \end{aligned} \tag{18}$$

For odd $m < K$ under $p^2 > m^{K-1}$,

$$\sup_{q\neq 0}|\mu_m(q)| \le \frac{2^{m/2+1}K^{m/2+1}}{p-1}. \tag{19}$$

Proof. Replace the balanced tuples of (13) by matching assignments. After division by 2^rK^{r-1}, (15) gives the third term of (18). Each noncrossing matching contributes exactly one; together they give $pC_r/(p-1)$, whose difference from C_r is the first term. For each crossing average over its 2^r orientations, apply (17), and apply the L^2 triangle inequality over the $P_r - C_r$ matchings. This is the second term. The last term is (14). Lemma 3 and (14) give (19). □

Corollary 5 (global law only). Suppose $K \to \infty$ through odd primes, $p \equiv 1 \pmod{K}$ is prime, and $K = o(\log p)$. Every fixed moment of the empirical spectral measure of X_q converges in L^2 over nonzero q to the standard semicircle moment. The empirical measures converge weakly in probability to

$$\rho_{\mathrm{sc}}(x)\,dx = \frac{\sqrt{4-x^2}}{2\pi}\mathbf{1}_{[-2,2]}(x)\,dx. \tag{20}$$

The averaged empirical measures converge weakly as well.

Proof. For every fixed m, eventually $m < K$ and (10) holds, while every fixed power of K divided by p tends to zero. Equations (18)–(19) give the moment limits. From any subsequence, select a further subsequence on which all integer moments converge almost surely, by diagonal extraction. Bounded second moments give tightness; bounded higher even moments give uniform integrability for each lower moment. Every weak subsequential limit therefore has the semicircle moments. Since $C_r \le 4^r$, any measure with these moments is supported on $[-2,2]$; polynomial approximation on that compact interval proves uniqueness. The subsequence criterion gives weak convergence in probability. Bounded continuous tests then also converge in expectation, proving the averaged assertion. □

This does not assert eigenvalue rigidity, an edge law, microscopic spacing statistics, or estimates uniform for m growing with n.

9. Obstruction for one fixed cat map and remaining work

Let $T \in \mathrm{SL}_2(\mathbb{Z})$ be one fixed hyperbolic matrix with order K modulo p. Then $T^K - I$ vanishes modulo p. At least one of its integer entries is nonzero, and its absolute value is at most $C_T\Lambda_T^K$, for constants depending only on T. Hence

$$p \le C_T\Lambda_T^K, \qquad \log p \le K\log\Lambda_T + \log C_T. \tag{21}$$

Thus $K = o(\log p)$ is impossible along such an order sequence. On the split semisimple locus the eigenvalue order equals the matrix order, so that interpretation does not avoid the obstruction.

The new global law is a theorem about an arithmetic matrix-design regime, not a fixed-cat-map BGS theorem. The exact trace formula, antiunitary, and orbit assembly remain valid outside the cutoff. The next substantive step is to control non-balanced reciprocal words in polynomial regimes rather than eliminate them with an exponential norm bound. Microscopic BGS then still requires a specified desymmetrized propagator and Heisenberg-scale joint-trace estimates. Neither assertion is proved here.

10. Verification and related work

The maintained regression checks 14 integer histograms against a separate C++ affine-fixed-point implementation, 32,054 oriented pairings through $r = 5$, three matrix representations, Gauss rephasing, and finite reciprocal cutoffs. Floating matrix agreement and exact finite certificates are labeled separately. The Lean module records small finite trace certificates only; it is not a formalization of Theorem 4 or Corollary 5.

Semicircle laws in finite harmonic analysis are not new in general. Gurevich and Hadani study normalized Gram fluctuations of incoherent dictionaries (arXiv:0812.2602). Forey, Fresan, Kowalski, and Wigderson construct arithmetic graph families with global semicircle spectra (arXiv:2511.21351) and introduce Jacobian graphs (arXiv:2603.13198). Rosenzweig studies fluctuations of cat-map

matrix elements (arXiv:0909.1410). These are relevant comparison objects, not unverified inputs to the proof above. The source audit does not establish priority for this particular compressed-observable theorem.